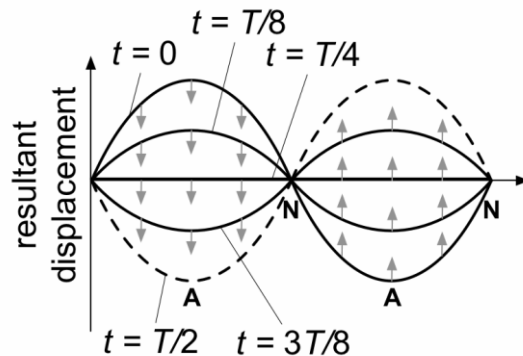


17 Which of the following statements about two consecutive antinodes in a stationary wave is false?

- A Their oscillations are in phase with each other.
- B Both particles at the antinodes will pass through their equilibrium positions with same speed.
- C The acceleration of the particles are always equal in magnitude and opposite in direction.
- D Both have maximum energies since the amplitudes of oscillations are maximum.
- A Points between two adjacent nodes are in phase. Points in adjacent segments are anti-phase.

All the points between *adjacent* nodes are **in phase** though they **do not have the same amplitude**. Points in adjacent segments are **anti-phase**.



18 Two protons are 1.0×10^{-14} m apart